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How to use the NTAG I²C *plus* for bidirectional communication

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Info	Content
Keywords	NTAG I²C *plus*, pass-through mode, bidirectional communication, SRAM
Abstract	This document describes the various data communication modes the NTAG I²C *plus* provides. Especially it is about the SRAM-based pass-through mode and also how to synchronize data transfer through the NTAG I²C *plus* in general.

Revision history

Rev	Date	Description
1.1	20180605	Editorial update and clarification on I²C address added
1.0	20160201	First release

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1. Introduction

The NTAG I²C *plus* provides with its wireless NFC and wired I²C interface the possibility for devices to get an easy communication channel to NFC devices.

The following sections describe:

- SRAM based pass-through mode
  The SRAM can be used for streaming data through the NTAG I²C *plus*, e.g. for data download or firmware updates. This is the mode where this document focuses on.
- Triggering the µC on NFC communication events
  The µC can be notified of various events on the RF interface to provide direct interaction when e.g. a NDEF message was read out completely
- Example pass-through transfer
  Here exemplary the command flow is shown how to do communication in both directions using the SRAM in pass-through mode

2. Pass-Through Mode

The pass-through mode allows the NTAG I²C *plus* to be used for bidirectional data transfer from a NFC device to an I²C bus master (e.g. a microcontroller). Use cases are for example:

- Read out of data collected in an embedded device (logging data)
- Upload new data in the embedded device (e.g. firmware update of the microcontroller)
- Bidirectional communication with exchange of commands and data (e.g. execute functions in the microcontroller or execute authentication schemes)

The pass-through mode provides the SRAM for data communication and triggering mechanisms for the synchronization of the data transfer.

3. Trigger Possibilities

The NTAG I²C *plus* has several triggering possibilities to provide input to a connected microcontroller to signal events on the RF interface. This signaling can be done through the Field Detection pin or through polling the equivalent registers over I²C. The field detection pin can react on the following events:

Table 1. Possible Settings of the FD pin

FD On (FD goes Low)	FD Off (FD goes high)
RF Field switched on	RF field switched off
Receive first valid start of communication	Receive HALT command or RF off
Tag is selected (anti-collision done complete)	Last NDEF Block read or RF off
pass-through mode – to make the FD triggering work you need to set both FD_ON and FD_OFF to this mode	

As an alternative to the FD pin also with the I²C register read commands the state of the tag can be checked. Important register bits which can provide information about the state of the tag are the following:

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Table 2. Register bits which can be used for communication synchronization

Register bit	Usage
PTHRU_ON_OFF	Detect if the pass-through mode is still enabled (gets reset in case of RF or I²C power down)
SRAM_MIRROR_ON_OFF	Detect if the SRAM Mirror Mode is still enabled (gets reset in case of I²C power down)
NDEF_DATA_READ	Detect if read was executed on the specified memory address. Is reset to 0 when the register was read
I2C_LOCKED	Detect if memory access is currently locked to I²C
RF_LOCKED	Detect if Memory access is currently locked to RF
SRAM_I2C_READY	Detect if there is data available in the SRAM buffer to be fetched by I²C
SRAM_RF_READY	Detect if there is data available in the SRAM buffer to be fetched by RF
RF_FIELD_PRESENT	Shows if a RF field strong enough to read the tag is there

Which registers need to be checked to create a pass-through communication is described in the following section.

4. Example Pass-Through Mode Transfer

Usually it is advisable that the I²C side configures the NTAG I²C *plus* to be able to receive data in the SRAM buffer. In this way the NFC device can write a request, a command or any data it wants to transfer to the SRAM. Then the embedded device decides upon this data if it should continue receiving data (leave the communication direction the same) or if the communication direction has to be changed to data transfer from I²C to NFC.

Thus, we first describe the data flow from NFC to I²C and then do a change in the communication direction.

4.1 Data Transfer NFC to I²C

Here the I²C side has first to wait until RF switches on the RF field. This can be done via monitoring the RF_FIELD_PRESENT bit in the register or via the FD pin.

If a RF field is detected, the pass-through direction is set and the pass-through mode is enabled from the I²C side.

The NFC side in the meanwhile can check if the pass-through mode is already switched on via polling on the bit “PTHROUGH_ON_OFF”. If this bit is “1b”, the pass-through mode is enabled.

Afterwards the NFC side immediately can start writing in the SRAM:

- NTAG I²C *plus* offers the FAST_WRITE command which allows writing the whole SRAM with one command. Normal WRITE command is also supported.
  NOTE: Previous product (NTAG I²C) offers only normal WRITE command.

When the last page in the SRAM was written, the NTAG I²C *plus* arbiter automatically switches the communication side. This means now:

- RF cannot access the memory (SRAM + EEPROM- get a NACK if it tries)
- I²C can now access the memory
- SRAM_I²C_READY gets set to 1b

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- If the FD pin is configured for pass-through mode (FD_ON and FD_OFF both set to 1b) then the FD pin gets activated (FD voltage is LOW)

The I²C side in the meanwhile either polls on the bit SRAM_I2C_READY to be set or waits on the FD pin to get low. Then the I²C side can fetch this data from the SRAM. On finish of read of the last block of the SRAM again the arbiter switched automatically the communication side:

- I²C cannot access the memory (SRAM + EEPROM, gets NACK if it tries)
- RF can now access the memory again
- SRAM_I2C_READY gets cleared
- If the FD pin is configured for pass-through mode (FD_ON and FD_OFF both set to 1b) then the FD pin gets deactivated (FD voltage is HIGH)

In case the NFC side should continue writing to the SRAM it can poll on the registers on the bit SRAM_I2C_READY to get unset or it simply tries writing to the memory and in case it gets a NACK it executes again the anti-collision to the tag and tries again.

In case now I²C should transmit data to the NFC side, first the pass-through mode needs to be turned off and then the communication direction can get changed.

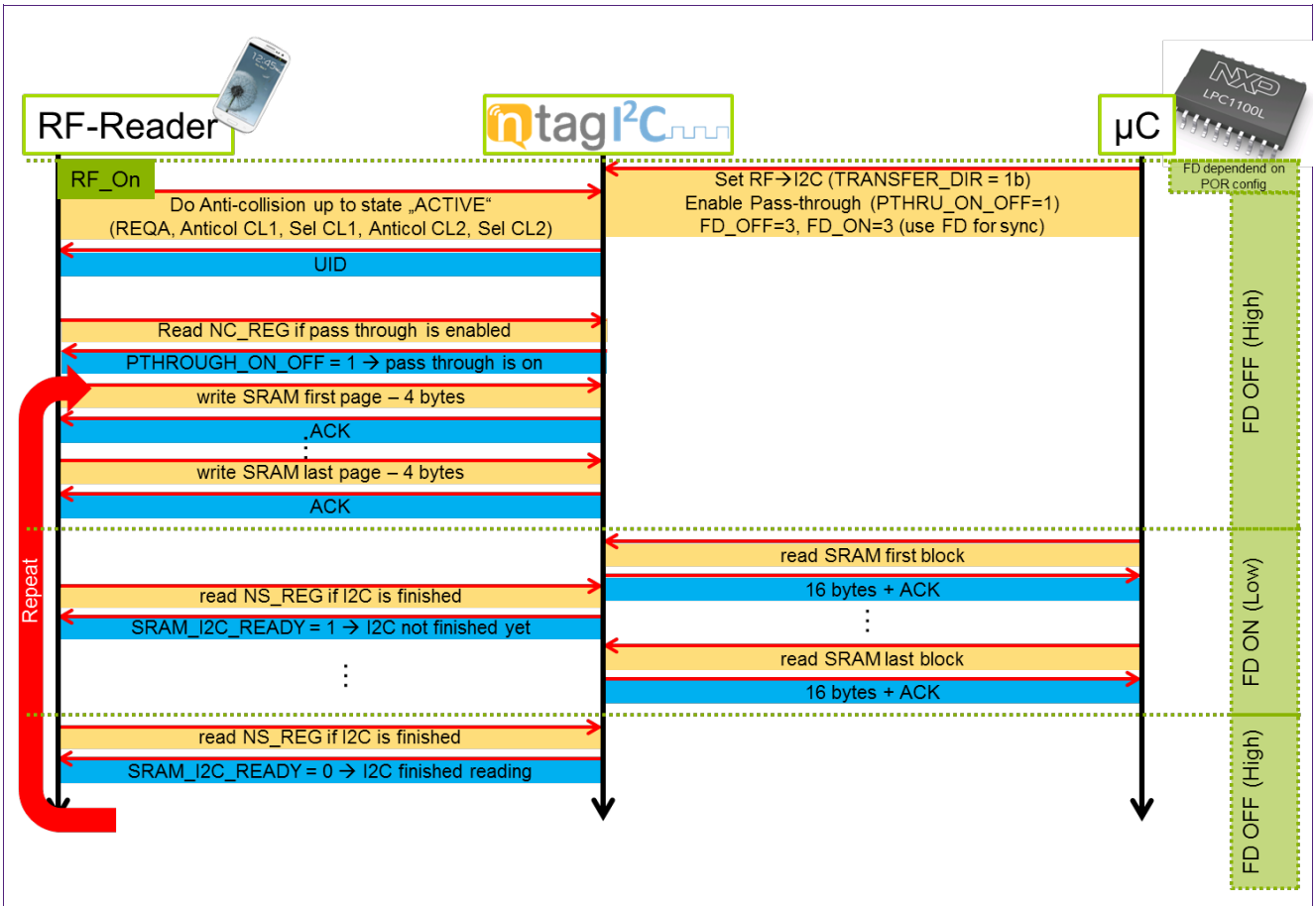


Fig 1. Block communication scheme of data transfer with pass-through from NFC to I²C

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4.2 Data Transfer from I²C to NFC

If I²C needs to transfer data to the NFC side, also here first both sides need to be powered. This can be done via monitoring the RF_FIELD_PRESENT bit in the register or via the FD pin.

If a RF field is detected, the pass-through direction is set and the pass-through mode is enabled from the I²C side.

In this communication direction the I²C side can immediately start writing data in the SRAM buffer. When the write of the last block of the SRAM buffer has finished, the NTAG I²C *plus* arbiter automatically switched the communication sides:

- I²C cannot access the memory (SRAM and EEPROM, gets NACK if it tries)
- RF can now access the memory again
- SRAM_RF_READY gets set
- If the FD pin is configured for pass-through mode (FD_ON and FD_OFF both set to 1b) then the FD pin gets deactivated (FD voltage is HIGH)

The NFC side in the meanwhile can check if data is already available in the SRAM via polling on the bit “SRAM_RF_READY” or it simply tries reading the memory and in case it gets a NACK it executes again the anti-collision to the tag and tries again.

Afterwards the NFC side can read the SRAM. This best is done using the “FAST_READ” command which allows reading the whole SRAM with one command. When the last page in the SRAM was read, the NTAG I²C *plus* arbiter automatically switches the communication side. This means now:

- RF cannot access the memory (SRAM and EEPROM- get a NACK if it tries)
- I²C can now access the memory
- SRAM_RF_READY gets reset
- If the FD pin is configured for pass-through mode (FD_ON and FD_OFF both set to 1b) then the FD pin gets activated (FD voltage is LOW)

The I²C side either can poll on the bit SRAM_RF_READY in the registers or listen on the FD pin to get low. Afterwards it can either write new data in the SRAM or switch the data flow direction via turning the pass-through mode off first, switch the direction and then turn it on again.

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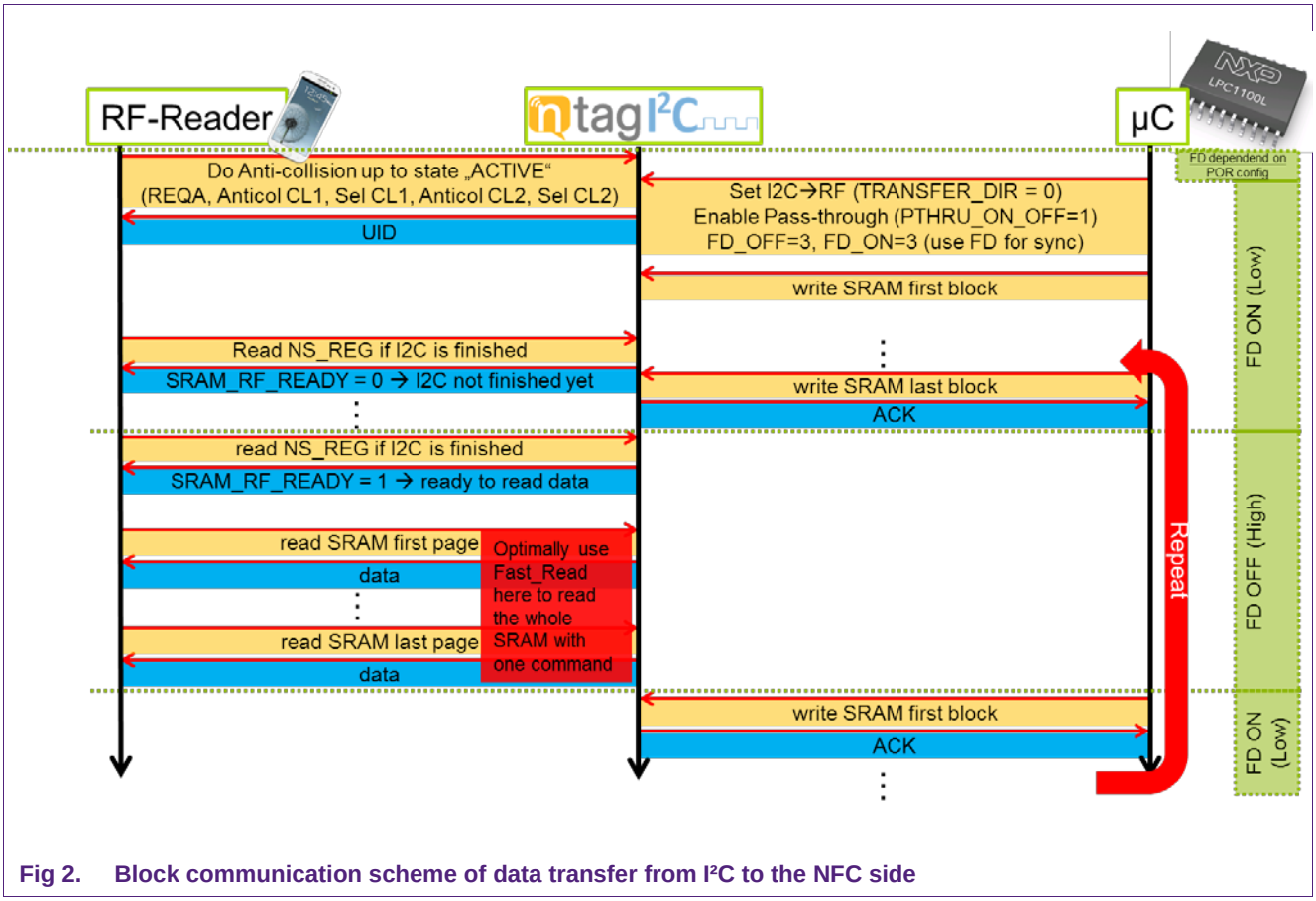


Fig 2. Block communication scheme of data transfer from I²C to the NFC side

4.3 Implementation hints

4.3.1 Register access for synchronization

The NTAG I²C *plus* allows due to the design with SRAM and registers in Sector 0 to choose for data flow synchronization between using the registers or the method to just try to write.

NOTE: Previous product (NTAG I²C) all the registers are only accessible for NFC in Sector 2 which raises the need to execute the SECTOR_SELECT command. This takes some time and some NFC devices don't fully support the SECTOR_SELECT command. So, on NTAG I²C it is often better to not use the registers for synchronization, but just wait some time and then execute the access again. For backward compatibility reasons these registers are duplicated to Sector 2 on NTAG I²C *plus*, however it is not recommended to use it for new designs.

4.4 Differences on the NTAG I²C and NTAG I²C *plus* for data communication

The exchange from NTAG I²C to NTAG I²C *plus* is package wise a drop-in replacement. From the SW point of view these differences have to be kept in mind:

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- GET_VERSION response
  To identify the four different types (memory size 1K and 2K, NTAG I²C and NTAG I²C *plus*), each NTAG I²C model has its own GET_VERSION response
- SRAM location as seen from NFC side
  The NTAG I²C *plus* has the SRAM now on the 1k and 2k memory size at the same location as the NTAG I²C 1k to enhance compatibility.
- Register access from NFC
  The set of registers on the NTAG I²C is on Sector 2, whereas in the NTAG I²C *plus* it is additionally duplicated in Sector 0. So, there is no need for a change, but the change is recommended as it increases compatibility.
- NDEF memory size
  User switching from the NTAG I²C 2k and using NDEF messages not fitting in the Sector 1 has to include the memory control TLV like described in the datasheet.
- Default Memory initialization
  The NTAG I²C *plus* is by default not NDEF formatted to increase the configuration flexibility. Before a NDEF message can be written to the tag the pages 03 and 04 have to be initialized like exemplary described in the datasheet.

  NOTE: When configuring Block 0 of NTAG I²C *plus* from I²C perspective, also I²C write address gets updated, because I²C write address is stored in the first byte of user memory. Reading out this first byte of NTAG I²C *plus*, it always returns 04h (UID0). Therefore, for convenience reasons, it is recommended to change default I²C device address 55h to 02h. As this first byte codes I²C write address, 04h (02h shifted left by one bit) value needs to be written to Byte 0.

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